Case Study: Texas Instruments SN74AUP1G04, Low-Power Single Inverter Gate

ECC-305 Microelectronics, Theory Case Study: commercially available IC / VLSI system (10 marks)

Student name / roll no. / batch / date: _____ / _____ / _____ / _____

Convention used in this report. [DS] marks a value or statement taken from the manufacturer's datasheet (SN74AUP1G04, document SCES571K, June 2004, revised June 2014, "Rev. K", retrieved from ti.com on 20 Oct 2026; TI's product page lists Rev. K as the current datasheet and the device status as ACTIVE). **[Expl]** marks my own explanation or calculation, not a manufacturer specification.

1. Device function and internal logic

[DS] The SN74AUP1G04 is a single inverter gate performing the Boolean function $Y = \overline{A}$ (function table: $A =$ H $\Rightarrow Y =$ L; $A =$ L $\Rightarrow Y =$ H). It belongs to TI's AUP ("Advanced Ultra-low-Power") family; per the datasheet the family has quiescent power consumption below 1 µA, and the device is offered in several packages from SOT-23 (DBV, 2.90 mm × 1.60 mm nominal) down to the 0.64 mm^2 X2SON (DPW, 0.8 mm × 0.8 mm) with 0.5 mm pitch. Pin functions (DBV/DCK/DRL): 1 N.C., 2 A (input), 3 GND, 4 Y (output), 5 V_{CC}.

[DS] Features that influence the internal design: balanced outputs; input hysteresis ($V_{hys} = 250$ mV typ. at 3.3 V, listed under Features) that tolerates slow input transitions; over-voltage-tolerant 3.6 V inputs/outputs; I_{off} circuitry for partial power-down and live insertion; and reduced drive strength that, in the datasheet's words, produces slower edges and prevents overshoot and undershoot.

[Expl] The datasheet gives only a functional block diagram (an inverter symbol); the internal transistor netlist is not published. At the logic level the function is that of a static CMOS inverter: a PMOS pull-up and an NMOS pull-down driven by the same input (Fig. 1). The real device also contains input hysteresis, I_{off}/back-drive protection and (probably) buffer stages, so the generic circuit is illustrative only. This is the same structure that the BBM paper (Case Study 2, ECC-353) modifies to remove short-circuit current.

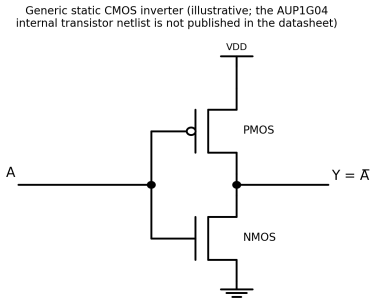


Figure 1. Generic static CMOS inverter (illustrative only; not TI's published internal netlist).

2. Supply-voltage range

Parameter **[DS]**	Value	Conditions / note
Recommended V_{CC}	0.8–3.6 V	Recommended operating conditions (optimised for 3.3 V per Features)
Absolute maximum V_{CC}	−0.5–4.6 V	Stress rating only
Input voltage V_I	0–3.6 V	Inputs 3.6 V tolerant at any valid V_{CC}; abs. max. −0.5–4.6 V
V_{IH} (min)	$0.65 \times V_{CC}$ (1.1–1.95 V); 1.6 V (2.3–2.7 V); 2 V (3–3.6 V)	High-level input voltage
V_{IL} (max)	$0.35 \times V_{CC}$ (1.1–1.95 V); 0.7 V (2.3–2.7 V); 0.9 V (3–3.6 V)	Low-level input voltage
Output drive I_{OH} / I_{OL} (max)	±20 µA (0.8 V); ±1.1 mA (1.1 V); ±1.7 mA (1.4 V); ±1.9 mA (1.65 V); ±3.1 mA (2.3 V); ±4 mA (3 V)	Recommended operating conditions
Input transition rate $\Delta t/\Delta v$	200 ns/V max	$V_{CC} = 0.8 - -3.6$ V
Operating temperature	−40–85 °C	Free-air

3. Power-related specifications

Parameter **[DS]**	Value	Test conditions
Static supply current I_{CC}	0.5 µA max. at 25 °C; 0.9 µA max. over −40–85 °C (the 0.9 µA figure is also quoted in Features)	V_I = GND or V_{CC}, $I_O = 0$, $V_{CC} = 0.8 - -3.6$ V
ΔI_{CC} (one input held below rail)	40 µA (25 °C) / 50 µA (−40–85 °C) max.	$V_I = V_{CC} - 0.6$ V, $I_O = 0$, $V_{CC} = 3.3$ V
I_{off} (partial power-down)	0.2 µA (25 °C) / 0.6 µA (−40–85 °C) max.	$V_{CC} = 0$ V, V_I or $V_O = 0 - -3.6$ V
Input leakage I_I	0.1 µA (25 °C) / 0.5 µA (−40–85 °C) max.	V_I = GND to 3.6 V, $V_{CC} = 0 - -3.6$ V
Power dissipation capacitance C_{pd}	3.9 pF typ. ($V_{CC} = 0.8 - -2.5$ V); 4.1 pF typ. ($V_{CC} = 3.3$ V ± 0.3 V)	$f = 10$ MHz, $T_A = 25$ °C
Input capacitance C_i / output C_o	1.5 pF typ. / 2.5 pF typ.	$V_I = V_{CC}$ or GND; C_o at $V_{CC} = 3.6$ V
Output levels	$V_{OH} \geq V_{CC} - 0.1$ V and $V_{OL} \leq 0.1$ V at $\lvert I_O \rvert =$ 20 µA (0.8–3.6 V); e.g. $V_{OH} \geq 1.32$ V (25 °C) at 1.65 V, $I_{OH} = -1.9$ mA	See datasheet Section 7.5
Thermal resistance $R_{\theta JA}$	298.6 °C/W (DBV) … 554.9 °C/W (DRY)	Thermal information table
ESD	HBM 2000 V, CDM 1000 V	Handling ratings; latch-up > 100 mA per JESD 78 Class II

Note on column assignment: the I_{CC}, ΔI_{CC}, I_{off} and I_I rows list only two numbers in the text extraction of the PDF; they are read here as ($T_A = 25$ °C max., −40–85 °C max.), which is consistent with the Features statement "$I_{CC} = 0.9$ µA max." I recommend confirming this against the PDF table if exact column placement matters.

One discrepancy turned up while checking. TI's product-page parametric table lists a maximum supply current of 10 µA, while the datasheet's electrical table gives $I_{CC} = 0.9$ µA max. (and ΔI_{CC} up to 50 µA). The parametric figure is probably defined under a different convention (for example, a worst case across the whole family); this report uses the datasheet value. The product page also gives the typical-delay descriptor "t_{pd} 5–10 ns", which is a family-level figure, not the specification at a given V_{CC} and load.

4. Propagation delay and loading conditions

[DS] Test conditions: load circuit of datasheet Figure 3 with $C_L = 5$, 10, 15 or 30 pF (including probe and jig), 1 M■ resistor to the output, measured at $V_M = V_{CC}/2$; input pulses from a generator with PRR $\leq$ 10 MHz, $Z_O = 50\,\Omega$, $t_r/t_f \leq 3$ ns; outputs measured one at a time, t_{PLH} and t_{PHL} are the same as t_{pd}.

t_{pd} A→Y **[DS]**	$T_A = 25\,°C$: min / typ / max (ns)	–40–85 °C: min / max (ns)
$C_L = 5$ pF, $V_{CC} = 1.8\,V \pm 0.15\,V$	2.2 / 3.4 / 5.9	1.4 / 7.0
$C_L = 5$ pF, $V_{CC} = 3.3\,V \pm 0.3\,V$	1.4 / 2.1 / 3.2	1.2 / 3.9
$C_L = 5$ pF, $V_{CC} = 1.2\,V \pm 0.1\,V$	3.3 / 5.9 / 10.8	2.1 / 13.5
$C_L = 10$ pF, $V_{CC} = 1.8\,V \pm 0.15\,V$	2.6 / 4.0 / 6.9	2.1 / 7.9
$C_L = 15$ pF, $V_{CC} = 1.8\,V \pm 0.15\,V$	3.2 / 4.6 / 7.4	2.6 / 8.5
$C_L = 30$ pF, $V_{CC} = 1.8\,V \pm 0.15\,V$	4.6 / 6.3 / 9.2	4.1 / 10.5
$C_L = 30$ pF, $V_{CC} = 3.3\,V \pm 0.3\,V$	3.2 / 4.0 / 5.4	2.9 / 6.2

Other values from the same tables: at $V_{CC} = 0.8\,V$ the datasheet gives a single t_{pd} value of 15.6 ns ($C_L = 5$ pF), 17.7 ns (10 pF), 19.5 ns (15 pF) and 25.4 ns (30 pF). Features: $t_{pd} = 3.9$ ns max. at 3.3 V (this is the $C_L = 5$ pF, –40–85 °C maximum).

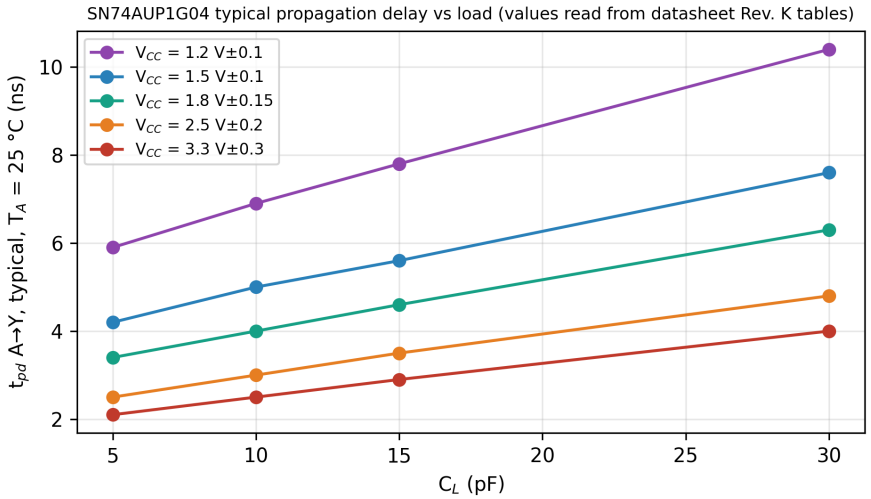


Figure 2. Typical propagation delay t_{pd} (A→Y, $T_A = 25\,°C$) of the SN74AUP1G04 versus load capacitance for five supply ranges; values read from the datasheet Rev. K tables.

Delay rises with load (about 0.5 ns per 5 pF near 1.8 V for the first steps, then more slowly per pF) and rises steeply as V_{CC} falls toward 0.8 V, because the drive current falls roughly with $(V_{CC} - V_T)^2$ **[Expl]**.

5. Worked example: dynamic versus static power

[Expl] (calculation using the datasheet C_{pd}, C_L and I_{CC}). Using the standard definition $P_{dyn} = (C_{pd} + C_L)\,V_{CC}^2 f$ (a general formula, not a datasheet statement) with $V_{CC} = 1.8$ V, $f = 10$ MHz, $C_{pd} = 3.9$ pF (datasheet) and $C_L = 5$ pF (the datasheet test load):

$$P_{dyn} = (3.9\,\text{pF} + 5\,\text{pF}) \times (1.8\,\text{V})^2 \times 10\,\text{MHz} = 8.9 \times 10^{-12} \times 3.24 \times 10^7\ \text{W} \approx 288\,\mu\text{W}.$$

The worst-case static power from the datasheet is $I_{CC}(\text{max}) \times V_{CC} = 0.9\,\mu\text{A} \times 1.8\,\text{V} \approx 1.6\,\mu\text{W}$, i.e. under 1 % of the dynamic power at 10 MHz. Dynamic power therefore dominates whenever the gate toggles; static power matters mainly for rarely toggling, battery-backed nodes. These are calculations at one chosen operating point, not specifications.

6. Applications and engineering relevance

[DS] Listed applications include ATCA solutions, active noise cancellation, barcode scanners, blood-pressure monitors, CPAP machines, cable solutions, embedded PCs, field transmitters, fingerprint biometrics, HVAC, NAS, server motherboards/PSU, software-defined radio, TVs, video-communications systems, wireless data-access cards/headsets/keyboards/mice and X-ray equipment.

[DS] Section 10: the AUP family is positioned for battery-powered portable products; the 3.6 V-tolerant I/O allows down-translation, for example dropping 3.3 V peripheral drivers to the 0.8–1.2 V levels of newer microcontrollers.

[Expl] Typical uses: signal inversion, glue logic, clock/enable conditioning, level-compatible interfacing, and (as a building block) simple oscillators or delay elements when combined with R and C. Relevance to ECC-305: a real-world instance of low-power CMOS design, with small C_{pd} and I_{CC}, reduced drive and slow edges to cut dynamic power and ground bounce, hysteresis for slow inputs, and I_{off} for partial power-down. The BBM paper (PBL component) is about removing the short-circuit part of dynamic power in exactly this kind of inverter-based stage.

7. Limitations and design trade-offs

- **Speed versus power.** t_{pd} is a few ns even at 3.3 V (3.2 ns max. at 25 °C, 3.9 ns over temperature, $C_L = 5\,\text{pF}$) and 10–15 ns at 1.2–0.8 V; the datasheet does not quote a maximum clock frequency.
- **Drive strength.** Only $\pm 4\,\text{mA}$ at 3 V, $\pm 1.9\,\text{mA}$ at 1.65 V and $\pm 20\,\mu\text{A}$ at 0.8 V; unsuitable for driving LEDs, long unterminated lines or heavy capacitive loads (delay rises with C_L).
- **Input conditions.** Input rise/fall must not exceed 200 ns/V; unused inputs must be tied to V_{CC} or GND; an input held between rails (the ΔI_{CC} condition) can draw up to 50 µA at 3.3 V.
- **Absolute limits.** $V_{CC} \leq 4.6\,\text{V}$ (abs. max.), continuous output current $\pm 20\,\text{mA}$, V_{CC}/GND current $\pm 50\,\text{mA}$; ESD ratings are modest (HBM 2000 V, CDM 1000 V), so ESD handling is needed.
- **Thermal.** $R_{\theta JA} \approx 300 - -555\,^{\circ}\text{C/W}$ depending on package; irrelevant at µW power but limits sustained heavy drive.
- **Family trade-offs** (TI product page, Similar products; **[DS]**/TI page). SN74LVC1G04 offers 1.65–5.5 V operation and much higher drive (stated average 24 mA) at the cost of higher power; TI also lists the newer SN74UVP1G04 (0.65–3.6 V, preview) as a lower-voltage alternative.

8. Questions the audience may ask (technically accurate answers)

Q: What does $Y = \overline{A}$ mean, and is the output ever undefined?

A: It is a logical inversion. Provided V_I is within the recommended levels (V_{IL} max. / V_{IH} min. for the V_{CC} used) and V_{CC} is within 0.8–3.6 V, the output is a defined logic level; between V_{IL} and V_{IH} the output is not guaranteed, which is why slow or floating inputs must be avoided (hysteresis helps with slow edges).

Q: Why is dynamic power much larger than static power?

A: Static current is only leakage ($I_{CC} \leq 0.9\,\mu\text{A}$), whereas every transition charges and discharges C_{pd} and C_L: $P = (C_{pd} + C_L)V_{CC}^2 f$. At 10 MHz, 1.8 V and 5 pF this is about 288 µW versus under 2 µW static (calculation above).

Q: How does the load capacitance change the delay?

A: Datasheet at 1.8 V, 25 °C, typical: 3.4 ns (5 pF), 4.0 ns (10 pF), 4.6 ns (15 pF), 6.3 ns (30 pF). The delay increases roughly linearly with C_L because the weak output stage supplies limited current.

Q: Why does the speed drop at low V_{CC}?

A: Drive current falls strongly as V_{CC} approaches V_T; typical t_{pd} rises from 2.1 ns at 3.3 V to 5.9 ns at 1.2 V ($C_L = 5$ pF), and a single 15.6 ns value is quoted at 0.8 V.

Q: What is I_{off} for?

A: With $V_{CC} = 0$ V the I_{off} circuitry disables the outputs and limits leakage to $\leq 0.6\,\mu$A (V_I or V_O up to 3.6 V), preventing back-drive current, which allows live insertion and partial power-down.

Q: What is the purpose of input hysteresis?

A: The datasheet states $V_{hys} = 250$ mV typ. at 3.3 V; it gives better noise immunity and allows slow input transitions without output chatter. It does not remove the 200 ns/V limit.

Q: Can a 3.3 V signal drive this gate when $V_{CC} = 1.2$ V?

A: Yes: inputs and outputs are specified 3.6 V tolerant, so the gate can down-translate 3.3 V signals to its V_{CC} level (datasheet applications section). Output levels are then limited to V_{CC}.

Q: How is this related to break-before-make?

A: Any static CMOS inverter has a brief moment of PMOS and NMOS both conducting (short-circuit current). AUP devices reduce it through slower, controlled edges and low drive; the BBM technique in the research paper removes it structurally with time-skewed gate signals but costs extra transistors and wiring. TI does not state that the AUP1G04 uses BBM.

Q: Why choose this gate instead of SN74LVC1G04?

A: For battery-powered products where power and tiny packages matter and loads are small; LVC offers higher drive and 5.5 V operation but consumes more power.

Q: Which numbers should I quote if asked for "the" propagation delay?

A: Always give V_{CC}, C_L, temperature and whether it is min/typ/max: for example, 3.4 ns typ. at 1.8 V, 5 pF, 25 °C; 3.9 ns max. at 3.3 V over temperature.

References

[1] Texas Instruments, "SN74AUP1G04 Low-Power Single Inverter Gate," datasheet SCES571K, June 2004, revised June 2014 (Rev. K), https://www.ti.com/lit/gpn/SN74AUP1G04 (retrieved 20 Oct 2026; package addendum dated Oct 2025).

[2] Texas Instruments, SN74AUP1G04 product page, https://www.ti.com/product/SN74AUP1G04 (status ACTIVE; parametric data; similar products; retrieved 20 Oct 2026).

[3] J. Puhan, D. Raič, T. Tuma, and Á. Bűrmen, "Break-before-Make CMOS Inverter for Power-Efficient Delay Implementation," *The Scientific World Journal*, vol. 2014, Art. 349131, doi:10.1155/2014/349131.